



Effect of strain rate on ductile fracture initiation in advanced high strength steel sheets: Experiments and modeling



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ABSTRACT

Low, intermediate and high strain rate tensile experiments are carried out on flat smooth, notched and central-hole tensile specimens extracted from advanced high strength steel sheets. A split Hopkinson pressure bar testing system is used in conjunction with a load inversion device to perform the high strain rate tension experiments. Selected surface strains, as well as local displacements, are measured using high speed photography in conjunction with planar digital image correlation (video extensometer). Through thickness necking precedes fracture in all experiments. A hybrid experimental–numerical approach is therefore employed to determine the strain to fracture inside the neck. To obtain an accurate description of the local strain fields at very large deformations, a plasticity model with a Johnson–Cook type of rate and temperature-dependency and a combined Swift–Voce strain hardening law is used in conjunction with a non-associated anisotropic flow rule. The incremental change in temperature is computed using a strain rate dependent weighting function instead of solving the thermal field equations. The comparison of the computed and measured force–displacement curves and surface strain histories shows good agreement before and after the onset of necking. From each experiment, the loading path to fracture is determined describing the evolution of the equivalent plastic strain in terms of the stress triaxiality, Lode angle parameter, strain rate and temperature. An empirical extension of the stress-state dependent Hosford–Coulomb fracture initiation model is proposed to account for the effect of strain rate on the onset of ductile fracture. The model is subsequently calibrated and successfully validated using the results from fracture experiments on DP590 and TRIP780 steels.

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1. Introduction

Advanced high strength steels (AHSS) are widely used in automotive engineering. During stamping, as well as during accidental crash loading, fracture may initiate in AHSS after loading at strain rates as high as 1000/s. Localized necking usually precedes fracture and a significant amount of strain is accumulated inside the localized neck. The strains cannot be measured directly inside a localized neck and a hybrid experimental–numerical approach must be used which requires visco-plasticity models that are accurate at very large strains.

Existing visco-plasticity models can be classified into physics-based models and phenomenological/empirical models. The so-called physics-based models are usually inspired by the thermodynamics and kinetics of slip as summarized in the mono-

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graph by [Kocks et al. \(1975\)](#). The constitutive equations may be formulated at the slip system level and then used to compute the effective material response using a polycrystal plasticity approach (e.g. [Balasubramanian and Anand, 2002](#)). Alternatively, physics-based models may be directly formulated at the macroscopic level by substituting the mechanical variables at the slip system level (e.g. resolved shear stress, shearing rates on slip systems) by their macroscopic counterparts (e.g. von Mises equivalent stress, equivalent plastic strain rate). The [Zerilli–Armstrong model \(1987\)](#) is based on a thermal activation analysis of dislocation motion which results in two different constitutive equations for the macroscopic flow stress of FCC and BCC materials, respectively. A comparison of the performance of the physics-based constitutive models by [Bodner and Partom \(1975\)](#), [Zerilli and Armstrong \(1987\)](#) and [Khan and Huang \(1992\)](#) is included in the review of rate- and temperature-dependent models by [Liang and Khan \(1999\)](#). Good agreement of the predictions of the Khan–Huang–Liang plasticity model with the results from static and dynamic tension experiments has been reported for AHSS ([Khan et al., 2012a](#)) and Ti–6Al–4V alloy ([Khan et al., 2012b](#)). The [Rusinek–Klepaczko model \(2001\)](#) assumes the additive decomposition of the flow stress into an internal and an effective stress, with an Arrhenius type of law providing the strain rate and temperature dependence of the latter component. [Voyiadjis and Abed \(2005\)](#) present an extended Zerilli–Armstrong model which can accurately describe the adiabatic high strain rate response of both FCC and BCC materials for large strains. A rate- and temperature-dependent plasticity model embedded into a continuum damage mechanics framework ([Brüning, 2006](#)) has been applied by [Brüning and Gerke \(2011\)](#). A unified viscoplastic model accounting for the rate-dependent ductile-to-brittle transition during microcracking in polycrystalline materials has been proposed by [Shojaei et al. \(2013\)](#). The effect of strain rate on the plastic anisotropy of advanced high strength steel has been investigated by [Huh et al. \(2011\)](#). Their experiments for strain rates of up to 100/s on TRIP590 and DP780 steels indicated a decrease in plastic anisotropy with increasing strain rate.

Unlike the above models, the widely used [Johnson–Cook model \(1983\)](#) has no physical basis and is purely empirical. It postulates the multiplicative decomposition of the flow stress into three functions that depend on the strain, the strain rate and the temperature, respectively. Many studies in impact engineering demonstrate that the simple J–C plasticity model provides reasonable predictions of the temperature-dependent viscoplastic response of metals up to very high strains (e.g. [Johnson and Cook, 1983](#); [Johnson and Holmquist, 1988](#); [Clausen et al., 2004](#); [Smerd et al., 2005](#); [Verleysen et al., 2011](#); [Erice et al., 2012](#); [Kajberg and Sundin, 2013](#)). However, it shows serious limitations when an accurate description of the hardening at large strains is required. An important example is the prediction of the necking and post-necking response of sheet metal. Under static and isothermal loading conditions, the J–C model reduces to a power-law which is known to overestimate the strain hardening capacity of automotive steels (e.g. [Dunand and Mohr, 2010](#)). This shortcoming has been addressed by [Sung et al. \(2010\)](#) through a constitutive model that makes use of a temperature-dependent combination of power and saturation hardening.

The effect of stress state on the ductile fracture of advanced high strength steels has been studied extensively at low strain rates (e.g. [Sun et al., 2009](#); [Mohr et al., 2010](#); [Luo and Wierzbicki, 2010](#); [Kim et al., 2011](#); [Gruben et al., 2011](#); [Chung et al., 2011](#); [Lou et al., 2014](#); [Lian et al., 2013](#)). The reader is referred to [Lecarme et al. \(2011\)](#), [Malcher et al. \(2012\)](#) and [Lou et al. \(2014\)](#) for reviews of the current literature on ductile fracture of non-AHSS materials. Using a high speed hydraulic testing machine, [Huh et al. \(2008\)](#) performed uniaxial tensile tests of four different advanced high strength steels (TRIP600, TRIP800, DP600 and DP800) for pre-necking strain rates ranging from 0.003/s to 200/s. They report an increase of the fracture elongation (engineering strain over a length of 30 mm) for the TRIP600 steel, and an approximately rate-insensitive elongation for the other three materials. [Curtze et al. \(2009\)](#) performed static and intermediate strain rate tensile experiments on a servohydraulic testing machine and high strain rate experiments using a tensile Hopkinson bar system. In their experiments at room temperature, the elongation at fracture (over a length of 6 mm) for a DP600 steel and a TRIP700 steel were approximately the same irrespective of the strain rate. It is worth noting that necking preceded fracture in all of the above high strain rate tensile experiments, but the strain to fracture inside the localized neck has not been determined.

Estimates of the local strain at the onset of fracture in high strain rate experiments are available for bulk materials. The results from dynamic experiments on axisymmetric notched tensile specimens have been the basis for the formulation of the empirical Johnson–Cook failure model ([Johnson and Cook, 1985](#)). Unlike for flat specimens, the local strains at fracture in round specimens may be conveniently estimated from cross-section measurements in combination with FEA-based or analytical correction formulas (e.g. [Bridgman, 1952](#)). [Johnson and Cook \(1985\)](#) reported an increase in ductility as a function of the strain rate for OFHC copper, ARMCO iron and 4340 steel. A positive strain rate sensitivity of the strain to fracture has also been determined from static and dynamic notched tension experiments on aluminum 5083–H116 ([Clausen et al., 2004](#)) and martensitic stainless steel FV535 ([Erice et al., 2012](#)). [Khan and Liu \(2012a,b\)](#) incorporated the effect of strain rate and temperature into their stress-based ductile criterion. Their model captures the significant increase in ductility observed during high strain rate torsion experiments on aluminum 2024–T351.

In the present work, low, intermediate and high strain rate experiments are performed on flat specimens with a central hole and different notches. The strain to fracture is determined inside the neck using detailed finite element analysis in conjunction with digital image correlation. To obtain a highly accurate description of the post-necking response of specimens extracted from DP590 and TRIP780 steel sheets, the Johnson–Cook plasticity model is coupled with a combined Voce–Swift strain hardening function and a non-associated anisotropic flow rule. The loading paths up to the onset of fracture are identified from all experiments, including the evolution of stress triaxiality, Lode angle parameter, temperature, strain rate and equivalent plastic strain. Subsequently, an empirical extension of the Hosford–Coulomb fracture initiation model is proposed and validated to account for the effect of strain rate on ductile fracture.

2. Experimental techniques

2.1. Material and specimens

The tested material is a 1.4 mm thick dual phase steel (DP590) sheet provided by ArcelorMittal. Eight different types of specimens (Fig. 1) are extracted from these sheets:

- Uniaxial tension (UT) specimens: The UT-specimen for the low and intermediate strain rate experiments (Fig. 1a) features a 10 mm wide and 20 mm long gage section. This particular specimen shape with a short gage section had been initially chosen to be able to carry out compression tests on the same specimen (which ultimately has not been explored in the framework of the present study).
- Compact uniaxial tension specimens (Fig. 1b): This specimen is designed for high strain rate testing and features a 5 mm wide and 19 mm long gage section.
- Notched tensile specimens with a 20 mm wide gage section and circular cutouts. The specimens feature a notch radius of $R = 6.67$ mm (NT6) and $R = 20$ mm (NT20), respectively, reducing the width of the gage section to 10 mm in the center.
- Specimens with a central hole (CH). The CH-specimens feature a 20 mm wide gage section with an 8 mm diameter hole in the center.

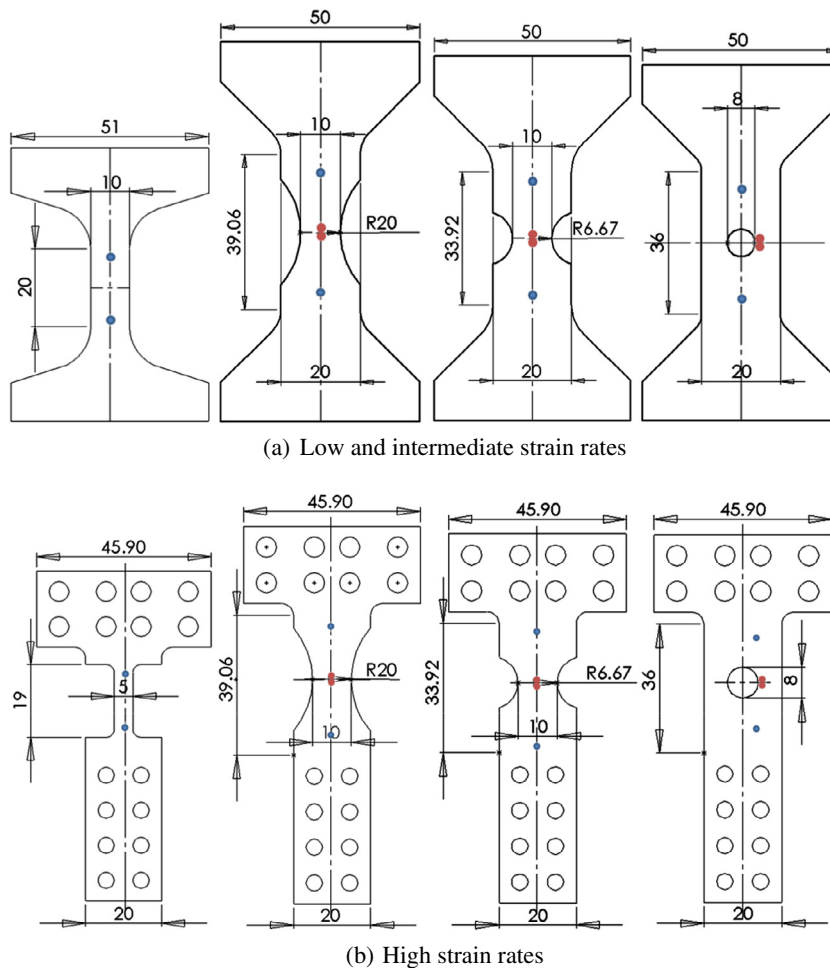


Fig. 1. Specimens for experiments at (a) low and intermediate strain rates, and (b) high strain rates. From left to right, each row shows the respective specimens for uniaxial tension (UT), notched tension (NT20), notched tension (NT6), and tension with a central hole (CH); blue solid dots highlight the position of the virtual extensometer for relative displacement and speed measurements; the red solid dots highlight the corresponding points for axial surface strain measurements. The blue dots for the CH specimens are positioned off the longitudinal axis to allow for the use of a narrow DIC window that includes both the red and blue dots. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Note that the same NT and CH gage section geometries are used for all strain rates. All specimens are extracted from the sheets using water jet cutting. The hole in the CH specimen is introduced through CNC milling to minimize the effect of the surface condition on the onset of fracture (see Dunand and Mohr, 2010). All specimens are prepared such that their tensile axis is aligned with the rolling direction.

Prior to testing, a random speckle pattern is applied to the specimens. The relative displacement of two points on the respective upper and lower specimen shoulders (see blue dots in Fig. 1) is measured in all experiments by means of digital image correlation (VIC2D, Correlated Solutions). The distance between these points is equal to $l_0 = 17.5$ mm for the UT-specimens, and $l_0 = 30$ mm for the NT and CH specimens.

2.2. Experimental technique for low strain rates

The experiments at low strain rates are carried out on a MTS hydraulic testing machine with a 250 kN load cell and custom-made high pressure clamps. The vertical actuator speeds for the low strain rate experiments are 2 mm/min (UT specimens) and 0.4 mm/min (NT and CH specimens), respectively. The software VicSnap is used in conjunction with a digital camera (AVT Pike F-505B/C) with a 90 mm 1:1 macro lens to acquire images at a frequency of 1 Hz with a square pixel edge length of approximately 27 μm at a resolution of 2452×2054 pixels.

2.3. Experimental technique for intermediate strain rates

The intermediate strain rate experiments are conducted on the same hydraulic testing machine as the low strain rate experiments. The UT-specimen is loaded at a cross-head speed of 20 mm/s, while the loading is applied at a constant rate of 8.3 mm/s for all other specimens. The images for DIC are acquired at a frequency of 2000 Hz using a high speed camera (Vision Research, Phantom 7.3) at a resolution of 800×600 pixels and a square pixel edge length of approximately 50 μm . The image acquisition is triggered by the rise in the force signal recorded by the load-cell.

2.4. Experimental set-up for high strain rates

All high strain rate experiments are carried out on a split Hopkinson pressure bar (SHPB) system equipped with a load inversion device for tensile testing (Dunand et al., 2013). Ordering the components from the left to the right, the testing system consists of a striker bar (1204 mm long, 20 mm diameter), an input bar (3021 mm long, 20 mm diameter), a load-inversion device and two output bars (2041 mm long, 14 mm diameter), with all bars made out of steel. The load-inversion device consists of a pusher and a stirrup (Fig. 2). The left shoulder of the specimen fits into a slot in the stirrup (matching the thickness of the specimen) and is clamped through eight M5 screws, while the right specimen shoulder is clamped onto the pusher. It is worth noting that the setup is designed such as to minimize the effects of bending waves due to the eccentricity of the input bar. To minimize spurious axial wave reflections, the cross section of the pusher is constant along its longitudinal axis and matches that of the input bar. Analogously, the cross-section of the stirrup matches that of the output bars. The assembly is guided through brass bearings with lubricated contact surfaces. A static pre-compression of about 500 N is applied to the system prior to dynamic loading to ensure proper contact of all components.

The history of the axial force $F[t]$ acting on the specimen is calculated based on the recorded output bar strain histories, $\epsilon_{tra,i}[t]$, using strain gages positioned at a distance of 299 mm from the stirrup/output bar interface,

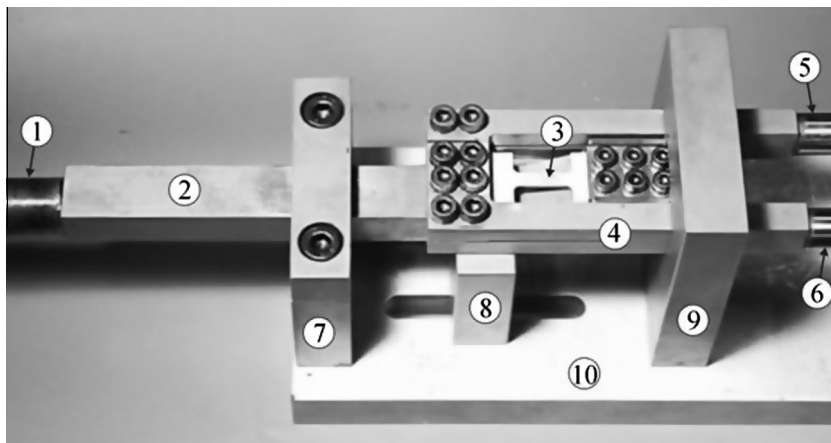


Fig. 2. Photograph of the load inversion device for high strain rate tensile testing (Dunand et al., 2013): (1) input bar ($\varnothing$ 20 mm), (2) pusher, (3) specimen, (4) stirrup, (5) first output bar ($\varnothing$ 14 mm), (6) second output bar ($\varnothing$ 14 mm), (7)–(9) guide blocks, and (10) base plate.

$$F_{out}[t] \cong -E_{out}A_{out} \sum_{i=1,2} \varepsilon_{tra,i}[t + \Delta t_s] \quad (1)$$

with the Young's modulus E_{out} , and the cross section area A_{out} of the output bar. Note that a time shift Δt_s needs to be introduced to account for the wave travel time from the specimen/stirrup interface to the stirrup/output bar interface.

For the displacement measurements, the same high speed camera as for the intermediate strain rate experiments (Vision Research, Phantom 7.3) is used. It is positioned at a distance of about 500 mm from the specimen which resulted in a square pixel edge length of about 57 μm . It is set to a resolution of 448×32 pixels and a frequency of 111 kHz (which corresponds to a 9 μs interval between subsequent images) for the UT experiments; a resolution of 512×32 pixels along with an acquisition frequency of 100 kHz is used for the NT and CH experiments.

3. Experimental results

3.1. Uniaxial tension

As shown by Mohr et al. (2010) for other AHSS materials, the true stress–strain curve for the DP590 steel obtained from static uniaxial tensile tests along different in-plane directions are almost identical (Fig. 3a), even though the corresponding Lankford ratios are direction dependent: $r_0 = 0.98$, $r_{45} = 0.81$, and $r_{90} = 1.13$. Assuming incompressibility, the Lankford ratios were determined from the slope of the logarithmic width strain versus axial strain curve for static loading conditions.

The effect of the strain rate on the engineering stress–strain curve for uniaxial tension along the rolling direction is illustrated by Fig. 3b. Only one curve is shown per loading velocity because of the excellent repeatability of all experiments. Table 1 provides a summary of the measured yield stresses, σ_{YS} , the ultimate strengths, σ_{UTS} , the engineering strains at ultimate

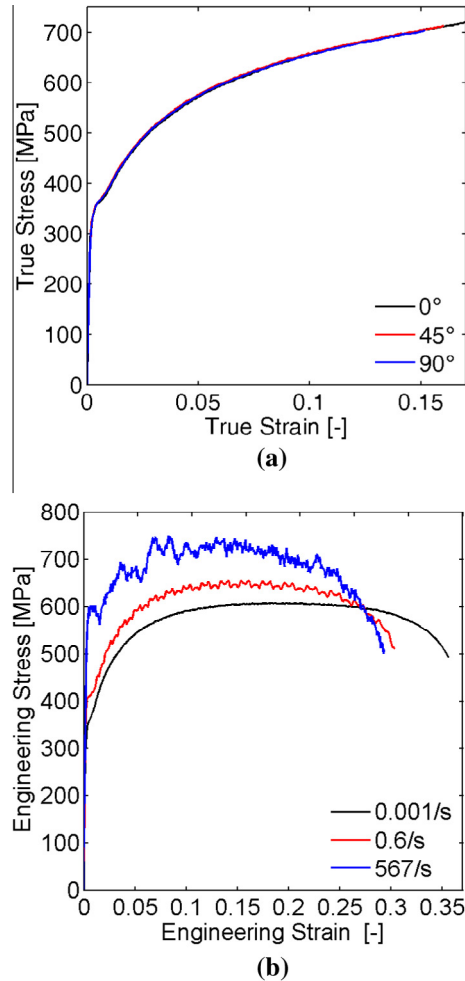


Fig. 3. (a) Plot of the true stress–strain curves as obtained from static uniaxial tension experiments with three different material orientations; (b) engineering stress–strain curves for low, intermediate and high strain rate experiments along the rolling direction.

strength, ε_U , and the engineering strains to fracture, ε_T . Observe that the material's yield stress is about 17% higher in the intermediate strain rate experiment than in the low strain rate experiment. It is also worth noting that the elongation to fracture (engineering strain at the onset of fracture as evaluated for an extensometer length of 17.5 mm) is significantly higher in the low strain rate experiment than in the intermediate and high strain rate experiments.

3.2. Fracture experiments

Fig. 4 shows representative force–displacement curves for the NT20, NT6 and CH experiments for three different speeds of loading (see Table 2); all curves are truncated at the apparent point of onset of fracture. For the low and intermediate strain rate experiments the onset of fracture may be determined either from a steep drop in the force level or a visible crack on the DIC image. In the high strain rate experiments, it is determined by the first visible crack on the specimen surface, since the recorded force history does not drop abruptly (which is attributed to the low wave speed of high frequency signals in bars).

All experiments show a force maximum prior to the onset of fracture, which increases as a function of the loading speed. The shapes of the force–displacement curves for low and intermediate strain rates are similar up to the force maximum, i.e. a dynamic increase factor could be used to describe the effect of the loading speed. The shape of the curve for the fast loading speed is different, i.e. it exhibits a significantly higher force level at the beginning of the experiment followed by a less pronounced hardening regime. Note that after the force maximum, the decrease in force is higher the faster the loading speed. However, it is remarkable to see that for all three specimen types, the displacement to fracture appears to be almost independent of the loading speed.

To gain further insight into the local deformation fields, we also report the strain determined from a 1 mm long virtual extensometer on the specimen surface (see secondary axis in Fig. 4). The extensometer is positioned on the longitudinal symmetry axis of the NT6 and NT20 specimens, and at a distance of 0.5 mm from the central hole boundary of the CH specimen (see pairs of red dots in Fig. 1). The evolution of the local strain as a function of the axial displacement is very similar irrespective of the speed of loading. In particular, only small variations are observed at the onset of fracture. For the NT20 specimens, the surface strain at the onset of fracture is 0.70 ± 0.03 . The corresponding strains for the NT6 and CH specimens are 0.55 ± 0.04 and 0.66 ± 0.03 , respectively. In addition to local strain measurements, post-mortem thickness profiles have been prepared from fractured NT20 specimens for all three loading speeds. The specimens are cut along their longitudinal axis of symmetry. The polished cross-sections are then photographed and measured using an optical microscope. The corresponding plots (Fig. 5a) shows the severe thickness reduction, from an initial thickness of 1.4 mm to less than 0.7 mm at the specimen center. Observe that the thickness reduction increases as a function of the speed of loading.

4. Rate-dependent plasticity model

An extension of the rate-independent non-associated quadratic plasticity model of Mohr et al. (2010) is proposed. The main purpose of the non-associated flow rule is to account for the direction dependency of the Lankford ratios in materials that exhibit a direction independent stress–strain response. Furthermore, Mohr et al. (2010) reported more realistic predictions of the strain localization when using a non-associated flow rule as compared to associated plastic flow. The proofs of the uniqueness of the stress distribution, the stability of plastic flow and the uniqueness of the stress and strain state are given by Stoughton (2002) for the rate-independent case. In close analogy with the Johnson–Cook plasticity model (Johnson and Cook, 1983), the deformation resistance is multiplicatively decomposed into a strain, strain rate and temperature dependent function. The incremental temperature increase is related to the strain rate and the rate of plastic work. As a result, the explicit dependence on the temperature is eliminated from the constitutive equations, while preserving the model's ability to account for thermal softening at high strain rates.

4.1. Yield function and flow rule

The yield function is described by the equation

$$f[\boldsymbol{\sigma}, k] = \bar{\sigma} - k = 0 \quad (2)$$

Table 1

DP590 material properties determined from the tensile tests: yield stresses, σ_{YS} , the ultimate strengths, σ_{UTS} , the engineering strains at ultimate strength, ε_U , and the engineering strains to fracture, ε_T .

$\dot{\varepsilon}$ [1/s]	0.001	0.001	0.001	0.63	70	143	270	567
α [°]	0	45	90	0	0	0	0	0
σ_{YS} [MPa]	352	353	353	412	523	565	588	599
σ_{UTS} [MPa]	607	607	605	655	*	*	732	740
ε_U [–]	0.19	0.18	0.17	0.16	*	*	0.13	0.14
ε_T [–]	0.36	0.33	0.32	0.30	*	*	**	0.29

* For strain rates of 70/s and 145/s the experiment was stopped before reaching the point of maximum force.

** For strain rate of 270/s experiment was stopped before fracture occurred.

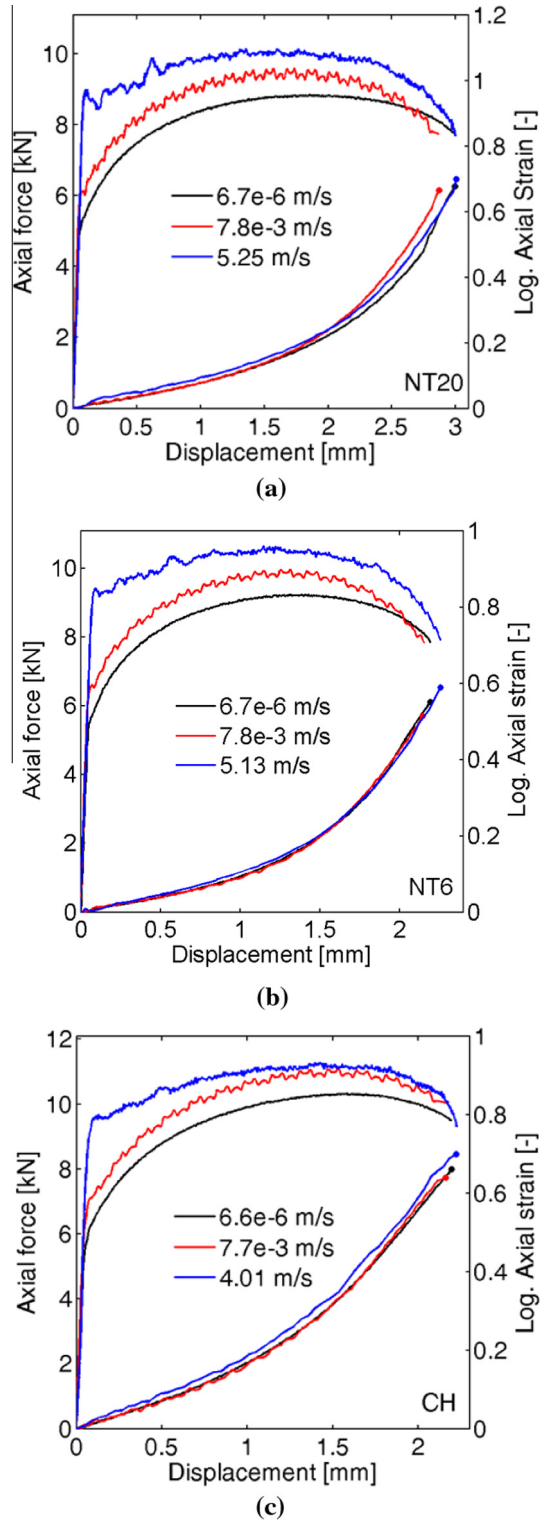


Fig. 4. Experimental results from fracture testing of the DP590 specimens: (a) NT20, (b) NT6, and (c) CH. The force–displacement curves are shown for different speeds of loading along with the corresponding axial strain evolutions on the secondary ordinate axis.

with the Mises equivalent stress $\bar{\sigma}$ and the deformation resistance k . The direction of plastic flow is aligned with the stress derivative of the flow potential function $g[\bar{\sigma}]$ (non-associated flow rule),

Table 2

Relative velocities of DP590 specimen shoulders as determined from measurements with a 30 mm long virtual extensometer. The corresponding values for the experiments on the TRIP780 specimens are given in parentheses.

	NT20 [m/s]	NT6 [m/s]	CH [m/s]
Slow	$6.7 \cdot 10^{-6}$ ($7.8 \cdot 10^{-6}$)	$6.7 \cdot 10^{-6}$ ($7.6 \cdot 10^{-6}$)	$6.6 \cdot 10^{-6}$ (n.a.)
Intermediate	$7.8 \cdot 10^{-3}$ ($7.8 \cdot 10^{-3}$)	$7.8 \cdot 10^{-3}$ ($7.8 \cdot 10^{-3}$)	$7.7 \cdot 10^{-3}$ (n.a.)
Fast	5.25 (7.1)	5.13 (4.2)	4.01 (n.a.)

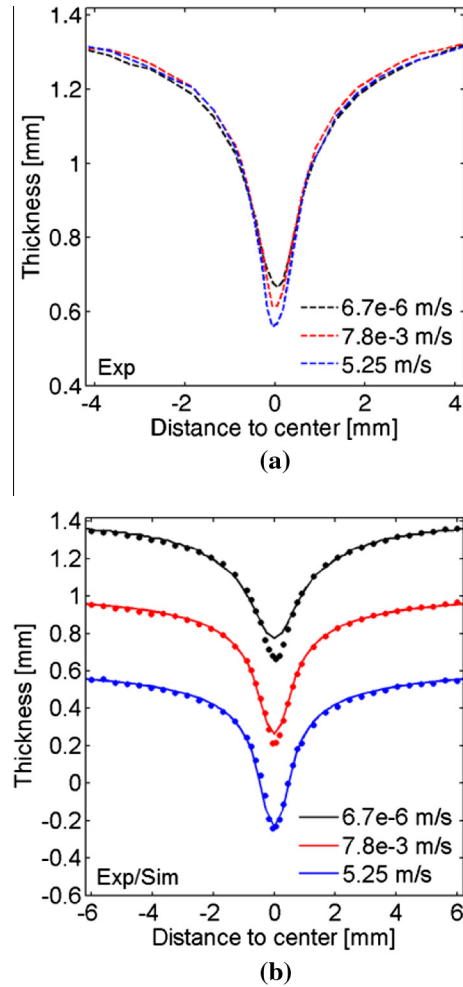


Fig. 5. (a) Measured thickness of a fractured NT20 specimen as a function of the distance along the longitudinal axis. (b) Superposition of the thickness profiles from experiments (dots) with those obtained from finite element analysis (solid lines). Intermediate and fast curves are shifted by -0.4 mm and -0.8 mm, respectively to improve readability.

Table 3

Lankford coefficients for DP590 and TRIP780.

	r_0 [-]	r_{45} [-]	r_{90} [-]
DP590	0.98	0.81	1.13
TRIP780	0.67	0.96	0.85

$$d\boldsymbol{\varepsilon}_p = d\lambda \frac{\partial g[\boldsymbol{\sigma}]}{\partial \boldsymbol{\sigma}} \quad (3)$$

$d\boldsymbol{\varepsilon}_p$ denotes the increment in the plastic strain vector (in the material coordinate system)

$$\boldsymbol{\varepsilon}_p = [\varepsilon_{11}^p \ \varepsilon_{22}^p \ \varepsilon_{33}^p \ 2\varepsilon_{12}^p \ 2\varepsilon_{23}^p \ 2\varepsilon_{13}^p]^T \quad (4)$$

$\boldsymbol{\sigma}$ is the Cauchy stress vector,

$$\boldsymbol{\sigma} = [\sigma_{11} \ \sigma_{22} \ \sigma_{33} \ \sigma_{12} \ \sigma_{23} \ \sigma_{13}]^T \quad (5)$$

and $d\lambda \geq 0$ is a scalar plastic multiplier. The flow potential is defined through the quadratic form

$$g[\boldsymbol{\sigma}] = \sqrt{(\mathbf{G}\boldsymbol{\sigma}) \cdot \boldsymbol{\sigma}} \quad (6)$$

with

$$\mathbf{G} = \begin{bmatrix} 1 & G_{12} & -(1 + G_{12}) & 0 & 0 & 0 \\ G_{12} & G_{22} & -(G_{22} + G_{12}) & 0 & 0 & 0 \\ -(1 + G_{12}) & -(G_{22} + G_{12}) & 1 + 2G_{12} + G_{22} & 0 & 0 & 0 \\ 0 & 0 & 0 & G_{33} & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 \end{bmatrix} \quad (7)$$

$g[\boldsymbol{\sigma}]$ actually corresponds to a special case of the Hill (1948) yield function which accounts for planar anisotropy with direction dependent Lankford ratios (see Table 3) through the anisotropy coefficients G_{12} , G_{22} and G_{33} . For $G_{12} = -0.5$, $G_{22} = 1$ and $G_{33} = 3$, the flow potential reduces to the isotropic von Mises equivalent stress.

The equivalent plastic strain increment $d\bar{\varepsilon}_p$ is defined as work-conjugate to the von Mises equivalent stress through the identity

$$\boldsymbol{\sigma} : d\boldsymbol{\varepsilon}_p = \bar{\sigma}(d\bar{\varepsilon}_p) \quad (8)$$

4.2. Decomposition of the deformation resistance

The Johnson–Cook model suggests the multiplicative decomposition of the deformation resistance into three functions that represent the effect of strain hardening, strain rate and temperature, respectively,

$$k[\bar{\varepsilon}_p, \dot{\bar{\varepsilon}}_p, T] = k_e[\bar{\varepsilon}_p] k_{\dot{\varepsilon}}[\dot{\bar{\varepsilon}}_p] k_T[T] \quad (9)$$

Inspired by the work of Sung et al. (2010), the strain hardening is represented by a linear combination of a power and exponential law,

$$k_e[\bar{\varepsilon}_p] = \alpha k_S[\bar{\varepsilon}_p] + (1 - \alpha) k_V[\bar{\varepsilon}_p] \quad (10)$$

with the power law (Swift, 1952)

$$k_S[\bar{\varepsilon}_p] = A(\bar{\varepsilon}_p + \varepsilon_0)^n \quad (11)$$

and the exponential law (Voce, 1948),

$$k_V[\bar{\varepsilon}_p] = k_0 + Q(1 - e^{-\beta \bar{\varepsilon}_p}) \quad (12)$$

The strain rate hardening term is represented by the logarithmic form

$$k_{\dot{\varepsilon}}[\dot{\bar{\varepsilon}}_p] = \begin{cases} 1 & \text{for } \dot{\bar{\varepsilon}}_p < \dot{\varepsilon}_0 \\ 1 + C \ln \left[\frac{\dot{\bar{\varepsilon}}_p}{\dot{\varepsilon}_0} \right] & \text{for } \dot{\bar{\varepsilon}}_p \geq \dot{\varepsilon}_0 \end{cases} \quad (13)$$

with the parameter C and the reference strain rate $\dot{\varepsilon}_0$. Furthermore, the standard Johnson–Cook term is used to account for thermal softening,

$$k_T[T] = \begin{cases} 1 & \text{for } T < T_r \\ 1 - \left(\frac{T - T_r}{T_m - T_r} \right)^m & \text{for } T_r \leq T \leq T_m \end{cases} \quad (14)$$

with the exponent m , the reference temperature T_r , and the melting temperature T_m of the tested material.

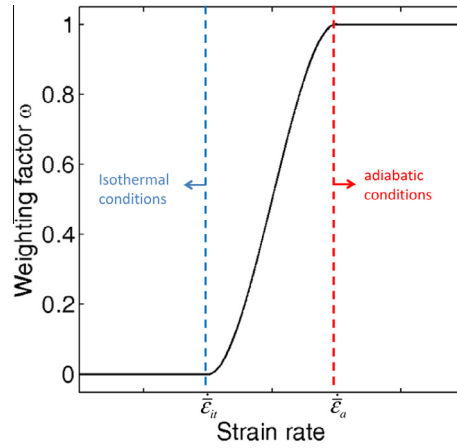


Fig. 6. Illustration of the weighting function ω defining the thermal conditions as a function of the strain rate.

4.3. Temperature evolution

The temperature is usually treated as an external state variable that is obtained from a full thermo-mechanical analysis (e.g. Sung et al., 2010). However, the computation of the temperature fields is often omitted in forming and crash simulations. This is not only due to the higher computational costs associated with thermo-mechanical analysis, but also due to significant uncertainties in the definition of thermal boundary conditions. For instance, in a forming simulation, the modeling of the heat flow between the sheet and the die is particularly difficult due to evolving contact surfaces.

The effect of thermal softening plays an important role in the post-necking range. It is worth noting that a temperature-independent Johnson–Cook model (i.e. setting $k_T[T] = 1$ in Eq. (9)) is not able to provide a reasonable approximation of the post-peak-load response of all our high strain rate experiments. To improve the predictions of purely mechanical analyses, we treat the temperature as an internal state variable which is determined through the evolution equation

$$dT = \omega[\dot{\epsilon}_p] \frac{\eta_k}{\rho C_p} \bar{\sigma} d\bar{\epsilon}_p \quad (15)$$

In (15), η_k quantifies the fraction of plastic work converted to heat (Taylor–Quinney coefficient¹), ρ is the mass density and C_p is the heat capacity. The strain rate dependent factor ω varies between 0 and 1. The limiting cases $\omega = 0$ and $\omega = 1$ correspond to isothermal conditions and adiabatic conditions, respectively. At low strain rates, the diffusion is typically much faster than the rate of heat generated which results in isothermal conditions ($\omega = 0$). At high strain rates, the opposite holds true: the diffusion is much slower than the rate of heat generated which results in adiabatic conditions ($\omega = 1$).

The temperature increase for intermediate strain rates is approximated in an ad hoc manner through the parametric form

$$\omega[\dot{\epsilon}_p] = \begin{cases} 0 & \text{for } \dot{\epsilon}_p < \dot{\epsilon}_{it} \\ \frac{(\dot{\epsilon}_p - \dot{\epsilon}_{it})^2 (3\dot{\epsilon}_a - 2\dot{\epsilon}_p - \dot{\epsilon}_{it})}{(\dot{\epsilon}_a - \dot{\epsilon}_{it})^3} & \text{for } \dot{\epsilon}_{it} \leq \dot{\epsilon}_p \leq \dot{\epsilon}_a \\ 1 & \text{for } \dot{\epsilon}_a < \dot{\epsilon}_p \end{cases} \quad (16)$$

with the parameters $\dot{\epsilon}_{it} > 0$ and $\dot{\epsilon}_a > \dot{\epsilon}_{it}$ defining the limits of the respective domains of isothermal and adiabatic conditions (Fig. 6). For ease of model calibration, the reference strain rate $\dot{\epsilon}_0$ of the rate-dependent J–C term is recommended as isothermal limit, i.e. $\dot{\epsilon}_{it} = \dot{\epsilon}_0$. Note that the above parametric form is actually the simplest form of a monotonic function over a finite domain that increases from 0 to 1 with zero slope at its end points. In the context of the present work, the main purpose of Eq. (16) is to act as a switch between isothermal and adiabatic conditions. Further research is needed to come up with a phenomenological (or at least empirical) function that provides an accurate description in the isothermal to adiabatic transition zone.

4.4. Plasticity model parameter summary

The constitutive model response is defined through 22 parameters:

¹ The assumption of a constant Taylor–Quinney coefficient does not apply to all metals. For example, in materials undergoing strain induced phase transformation, the effective Taylor–Quinney coefficient changes as the material deforms (Zaera et al., 2013). We neglect this effect when modeling TRIP-assisted steels in the sequel due to the small volume fraction of retained austenite and the expected completion of the phase transformation at moderate strains (<0.1).

- Two elasticity constants E and ν ;
- Seven parameters describing strain hardening: the Voce parameters $\{k_0, Q, \beta\}$, the Swift parameters $\{A, \varepsilon_0, n\}$, and the weighting factor α ;
- Three parameters $\{G_{12}, G_{22}, G_{33}\}$ specifying the anisotropic flow rule;
- Two parameters $\{C, \dot{\varepsilon}_0\}$ describing the effect of strain rate on plasticity;
- Three parameters $\{T_r, T_m, m\}$ describing the effect of temperature on plasticity;
- Four parameters determining the temperature evolution: the Taylor–Quinney coefficient η_k , the material density ρ and the heat capacity C_p ;
- Two model parameters describing the thermal conditions: the isothermal limit strain rate $\dot{\varepsilon}_{it}$ and the adiabatic limit strain rate $\dot{\varepsilon}_a$. These two parameters are part of the plasticity model, but these are not material parameters per se. They are determined by the loading/deformation and environmental conditions.

5. Finite element analysis

Finite element simulations are performed of all experiments to (1) facilitate the inverse identification of the material model parameters, to (2) validate the proposed plasticity model, and to (3) extract the so-called *loading path to fracture* for different stress states and strain rates.

5.1. FE meshes and boundary conditions

All simulations are carried out as dynamic explicit analyses with the finite element program [Abaqus \(2011\)](#). Uniform mass scaling is applied in low and intermediate strain rate simulations, such that the simulations terminate after approximately 200,000 time steps. The high strain rate simulations are performed without any mass scaling.

Due to the symmetry of the mechanical systems, only one eighth of each specimen (NT6, NT20, CH) is modeled ([Fig. 7](#)). The height of each mesh equates to half the length of the DIC extensometer from the respective experiment (recall blue dots in [Fig. 1](#)). The measured displacement history is applied to all nodes on the upper faces of the FE meshes. A zero normal displacement constraint is applied to the nodes positioned on the three symmetry planes of the specimens. Following the recommendation of [Dunand and Mohr \(2010\)](#), a mesh with an element length of $l_e = 100 \mu\text{m}$ and eight first-order solid elements over half the thickness is utilized to model through-thickness necking. Note that quasi-static loading conditions prevail in all experiments (see [Dunand et al., 2013](#)). As a consequence, the modeling of the full specimen (or even the entire SHPB system) could be omitted.

5.2. Plasticity model parameter identification

The plasticity model parameters are identified as follows:

- The anisotropy parameters are determined directly from the Lankford ratio measurements (at low strain rates) using the analytical relationships

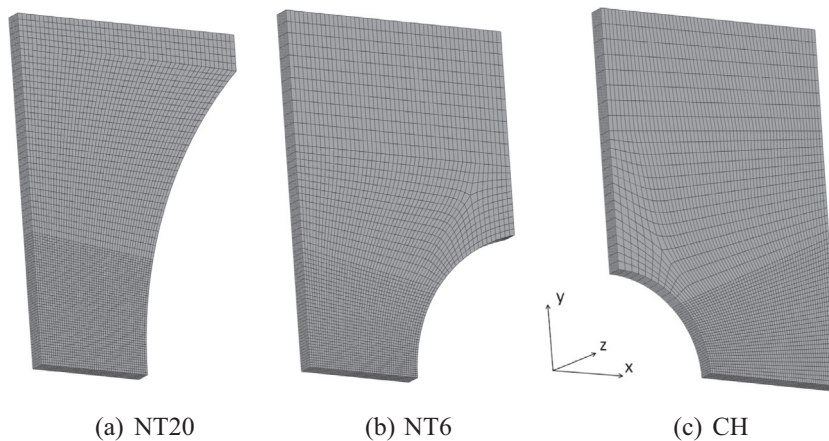


Fig. 7. Finite element models of the fracture specimens. Each model has eight elements through half the thickness and an element edge length of $l_e = 100 \mu\text{m}$ in the vicinity of the assumed location of onset of fracture.

$$G_{12} = -\frac{r_0}{1+r_0}, \quad G_{22} = \frac{r_0}{r_{90}} \frac{1+r_{90}}{1+r_0} \quad \text{and} \quad G_{33} = \frac{1+2r_{45}}{r_{90}} \frac{r_0+r_{90}}{1+r_0} \quad (17)$$

- The Voce and Swift strain hardening parameters are obtained from a least-square fit of the true stress-true strain curve obtained from low strain rate tension tests (up to the onset of necking).
- The room temperature is chosen as reference temperature $T_r = 293$ K.
- The parameters $\{E, \nu, \rho, c_p, \eta_k\}$ are taken from the literature (see Table 4).
- The remaining parameters $\mathbf{p} = \{\alpha, C, \dot{\epsilon}_0, T_m, m, \dot{\epsilon}_a\}$ are identified through inverse analysis of the NT20 experiments (which includes the post-necking range).

After simulating the NT20 experiment for low, intermediate and high speed loading, for a given parameter set $\mathbf{p}_k$, the difference between the experimental and computed force–displacement curves is quantified through the cost function

$$\Phi[\mathbf{p}_k] = \sum_{j=1}^3 \frac{1}{N_j} \sum_{i=1}^{N_j} w[u]_{j,i} (F_{j,i}^{\text{exp}}[u] - F_{n,i}^{\text{sim}}[u, \mathbf{p}_k])^2 \quad (18)$$

Here F_i^{exp} and F_i^{sim} are the force levels at the i th experimental data point (with a total of $N_j = 100$ points per curve). The index j denotes the respective experiment which is evaluated (low, intermediate or high strain rate); $w[u]_{n,i}$ is a weighing factor that is chosen as $w(u \leq u[F_{\text{max}}]) = 1$ up to the force maximum and $w(u > u[F_{\text{max}}]) = 2$ from the force maximum to the point of fracture. Emphasis is put on the second part of the load–displacement curve, as most plastic strain accumulates after the force maximum. It is of great importance for the reliability of the fracture strain estimates that the simulated force–displacement curve matches the experimental one in this range.

A minimum of the cost function in the five-dimensional parameter space is sought using a derivative-free simplex algorithm (Nelder and Mead, 1965). A solution satisfying $\phi < 0.1N^2$ (Table 4) is obtained after 264 iterations. To illustrate the behavior of the calibrated constitutive model, we plotted its stress–strain response for uniaxial tension at different strain rates in Fig. 8a. To elucidate the competition of the effects of strain rate and temperature, we computed also the model response for the theoretical limiting cases of isothermal conditions ($\omega = 0$) and adiabatic conditions ($\omega = 1$). The comparison of Fig. 8a with Fig. 8b and c shows that the rate effect is dominant at small strains (up to about 0.2), before thermal softening dominates the hardening response at intermediate and high strain rates.

5.3. Comparison of the experiments and simulations

Figs. 9–11 compare the force–displacement and axial strain–displacement curves from the simulations (solid lines) with the experimental results (dotted lines) for all tested specimen geometries and all speeds of loading. Recall that the NT20 experiments (Fig. 9) have been used to calibrate the model. The good agreement of the measured and simulated curves for NT6 (Fig. 10) and CH specimens (Fig. 11) may therefore be seen as validation of the proposed plasticity model.

- For the NT20 experiments at slow and intermediate speeds of loading (Fig. 9a and c), the computed force–displacement curves (solid black lines) match those measured experimentally (dotted black curves) very well. The overall prediction of the surface strain evolution (blue curves) is also satisfactory. For the high speed simulation (Fig. 9e), the force–displacement curves also agree well, except for the onset of inelastic deformation, where the experiment exhibits a higher force than the simulation.
- For NT6 (Fig. 10), similar observations are made as for NT20, i.e. good agreement of experiment and simulations is observed for both the force–displacement and strain–displacement curves, except for the initial force level at high velocity loading.
- For the slow and intermediate speed CH experiments, the simulation results agree well with the experimental curves (Fig. 11a and c). For the fast CH simulation (Fig. 11c), the force level at peak load is exceeded by about 10%, but the evolution of the axial strain is captured well by the simulation.

Table 4

Plasticity model parameters for DP590; the corresponding values for the TRIP780 steel are given in parentheses.

A [MPa]	ϵ_0 [–]	n [–]	k_0 [MPa]	Q [MPa]	β [–]
1031.0 (1526.5)	0.00128 (0.00519)	0.199 (0.242)	350 (488.91)	324.25 (447.5)	25.92 (18.65)
α [–]	C [–]	$\dot{\epsilon}_0$ [1/s]	T_r [K]	T_m [K]	m [–]
0.640 (0.536)	0.01366 (0.00557)	0.00116 (0.00321)	293 (293)	1673.7 (1806.7)	0.921 (1.025)
E [GPa]	ν [–]	C_p [J/kgK]	ρ [kg/m ³]	$\dot{\epsilon}_a$ [1/s]	η_k [–]
210 (185)	0.33 (0.33)	420 (420)	7850 (7850)	1.379 (1.343)	0.9 (0.9)

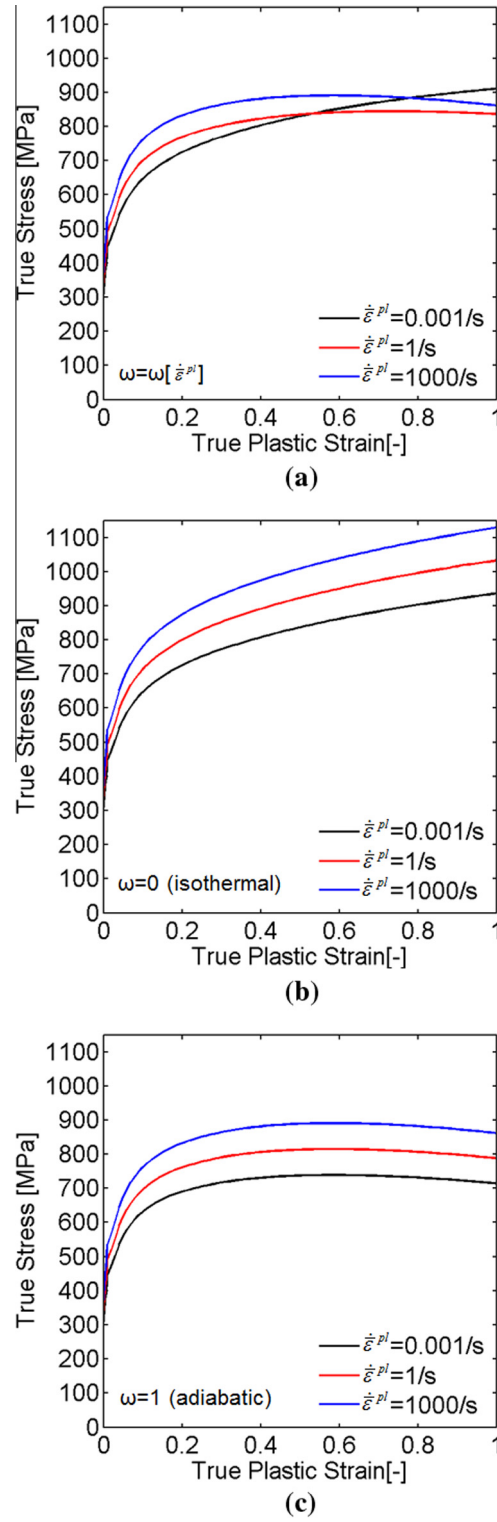


Fig. 8. Effect of strain rate on stress–strain response of the calibrated plasticity model (DP590) for (a) variable thermal conditions ($\omega = \omega[\dot{\epsilon}^{pl}]$), (b) for isothermal conditions ($\omega = 0$), (c) for adiabatic conditions ($\omega = 1$).

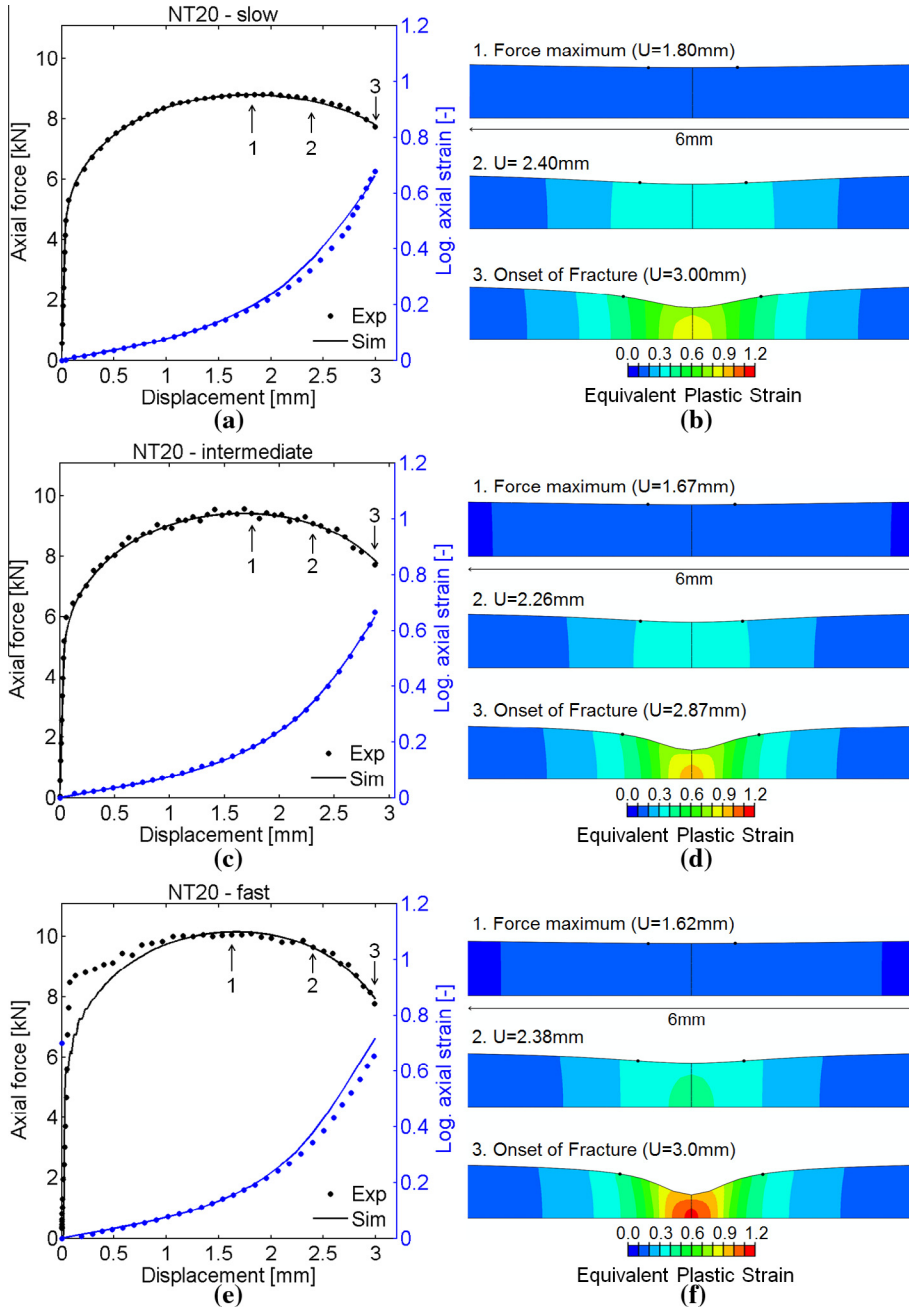


Fig. 9. Force–displacement and surface strain versus displacement curves from the experiments (points) and the simulations (lines) at different strain rates for the NT20-specimen. The right columns show selected contour plots of the equivalent plastic strain distribution within a longitudinal section of the specimen. The black dots on the contour plots indicate the current position of the surface strain extensometer. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

5.4. Loading paths to fracture

The term *loading path to fracture* is employed to refer to the evolution of the mechanical field and state variables at the location within the specimen where ductile fracture initiates. In particular, we are interested in the evolution of the equivalent plastic strain $\bar{\epsilon}_p$ as a function of the strain rate $\dot{\epsilon}_p$, the temperature T , the stress triaxiality η and the Lode angle parameter θ . The latter two are defined through the ratios of the stress tensor invariants,

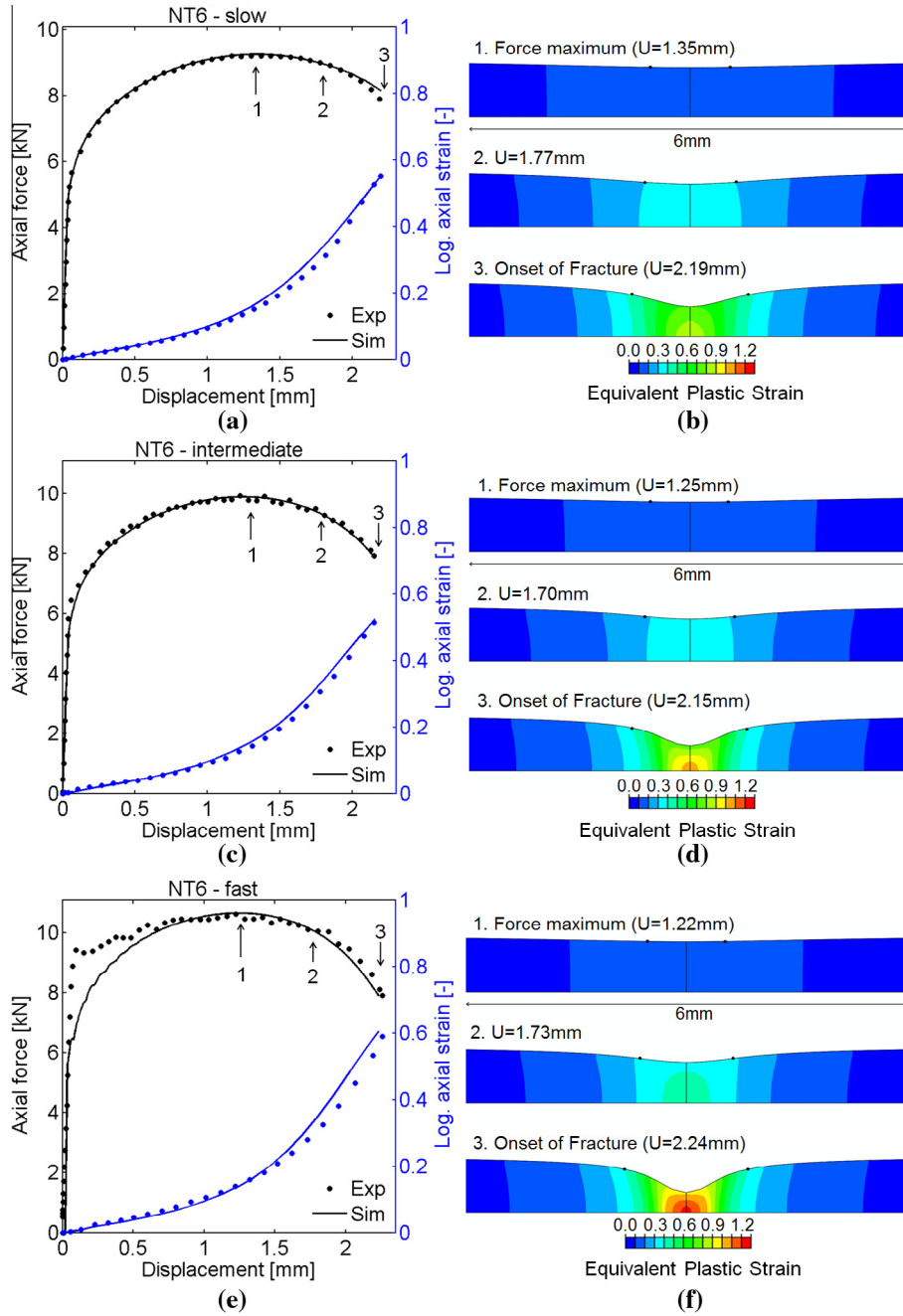


Fig. 10. Force–displacement and surface strain versus displacement curves from the experiments (points) and the simulations (lines) at different strain rates for the NT6-specimen. The right columns show selected contour plots of the equivalent plastic strain distribution within a longitudinal section of the specimen. The black dots on the contour plots indicate the current position of the surface strain extensometer.

$$\eta = \frac{I_1}{3\sqrt{3}J_2} \quad (19)$$

$$\bar{\theta} = 1 - \frac{2}{\pi} \arccos \left[\frac{3\sqrt{3}}{2} \frac{J_3}{(J_2)^{3/2}} \right] \quad (20)$$

with the invariant definitions

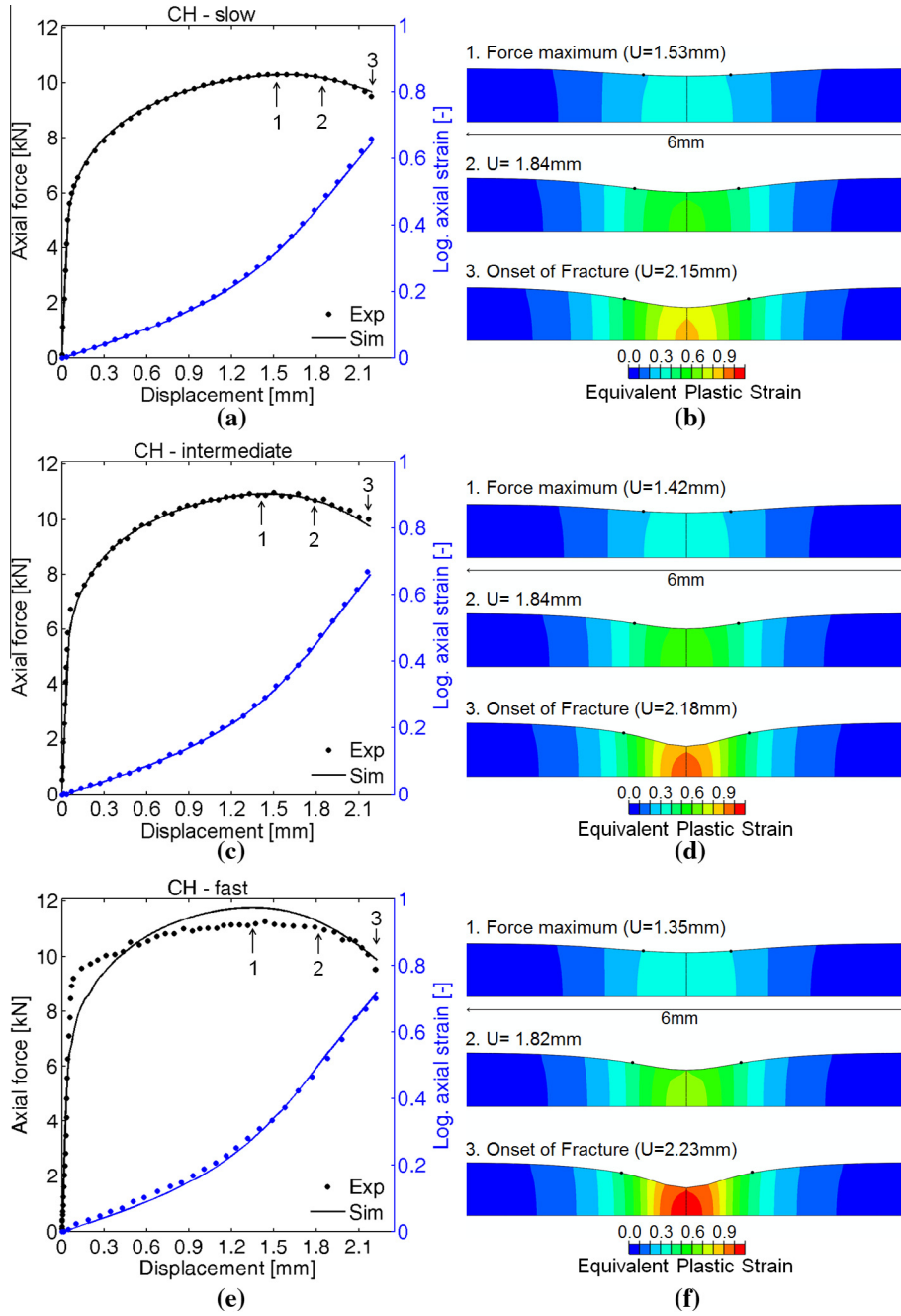


Fig. 11. Force–displacement and surface strain versus displacement curves from the experiments (points) and the simulations (lines) at different strain rates for the CH-specimen. The right columns show selected contour plots of the equivalent plastic strain distribution within a longitudinal section of the specimen. The black dots on the contour plots indicate the current position of the surface strain extensometer.

$$I_1 = \text{tr} \boldsymbol{\sigma}, \quad J_2 = \frac{1}{2} \mathbf{s} : \mathbf{s}, \quad J_3 = \det[\mathbf{s}] \quad \text{and} \quad \mathbf{s} = \text{dev}[\boldsymbol{\sigma}] \quad (21)$$

According to (20), the Lode angle parameter varies from $\bar{\theta} = -1$ (axisymmetric, generalized compression) to $\bar{\theta} = 1$ (axisymmetric, generalized tension), with $\bar{\theta} = 0$ for generalized shear.

One important assumption will be that fracture initiates at the integration point with the highest equivalent plastic strain at the instant of specimen fracture. For the NT20 and NT6 specimens, this point is positioned at the very center of the specimen, i.e. at the intersection point of all planes of specimen symmetry. In the CH specimens, this point is positioned on the specimen mid-plane at a distance of about 0.5 mm from the central hole boundary. This is due to the fact that the thickness

reduction is most pronounced at a small distance away from the hole boundary. Decreasing the hole diameter would move the point of highest equivalent plastic strain towards the hole boundary. However, the fracture initiation might then be affected by initial surface imperfections on the hole boundary due to machining.

Fig. 12 shows a summary of all loading paths to fracture in terms of the equivalent plastic strain, the stress triaxiality and the Lode angle parameter. The cross symbols on the loading paths indicate the point at which the force maximum is reached. Irrespective of the specimen type (NT20, NT6, CH), the stress state is independent of the strain rate and remains

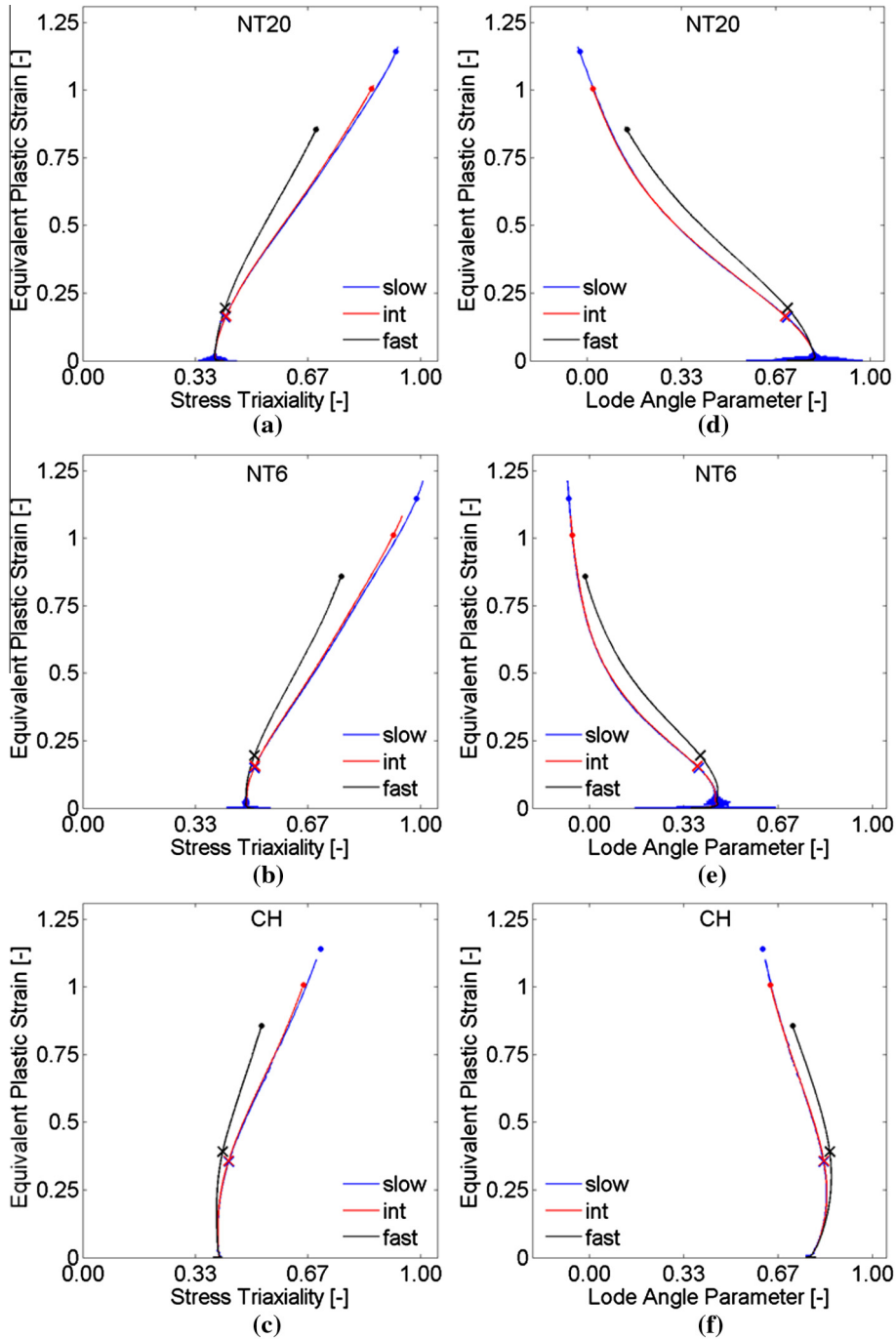


Fig. 12. Loading paths to fracture for NT20 (first line), NT6 (second line) and CH (third line) experiments on the DP590 steel. The left column figures show the equivalent plastic strain as a function of the stress triaxiality, the right columns show the evolution of the equivalent plastic strain as a function of the Lode angle. On each curve, the fracture strain predicted by the rate-modified Hosford–Coulomb model is indicated by a solid dot; the instant of maximum force is highlighted by a cross symbol.

approximately constant up to the force maximum. Thereafter, the stress triaxiality increases while the Lode angle parameter decreases. This is attributed to the built-up of an out-of-plane stress during through-thickness necking. Due to the absence of thermal softening for slow loading, necking is less pronounced for slow loading than for intermediate and fast loading. A plot of the temperature evolution (Fig. 13) reveals an increase in temperature at a rate of about $dT/d\bar{\epsilon} = 223$ K and $dT/d\bar{\epsilon} = 273$ K for the intermediate and fast experiment, respectively, while the temperature remains constant throughout the slow experiments (according to the model). The predicted temperature increase during an intermediate strain rate experiment has been confirmed through a thermocouple measurement on the specimen surface ($\Delta T = 80$ K). However, further research is needed

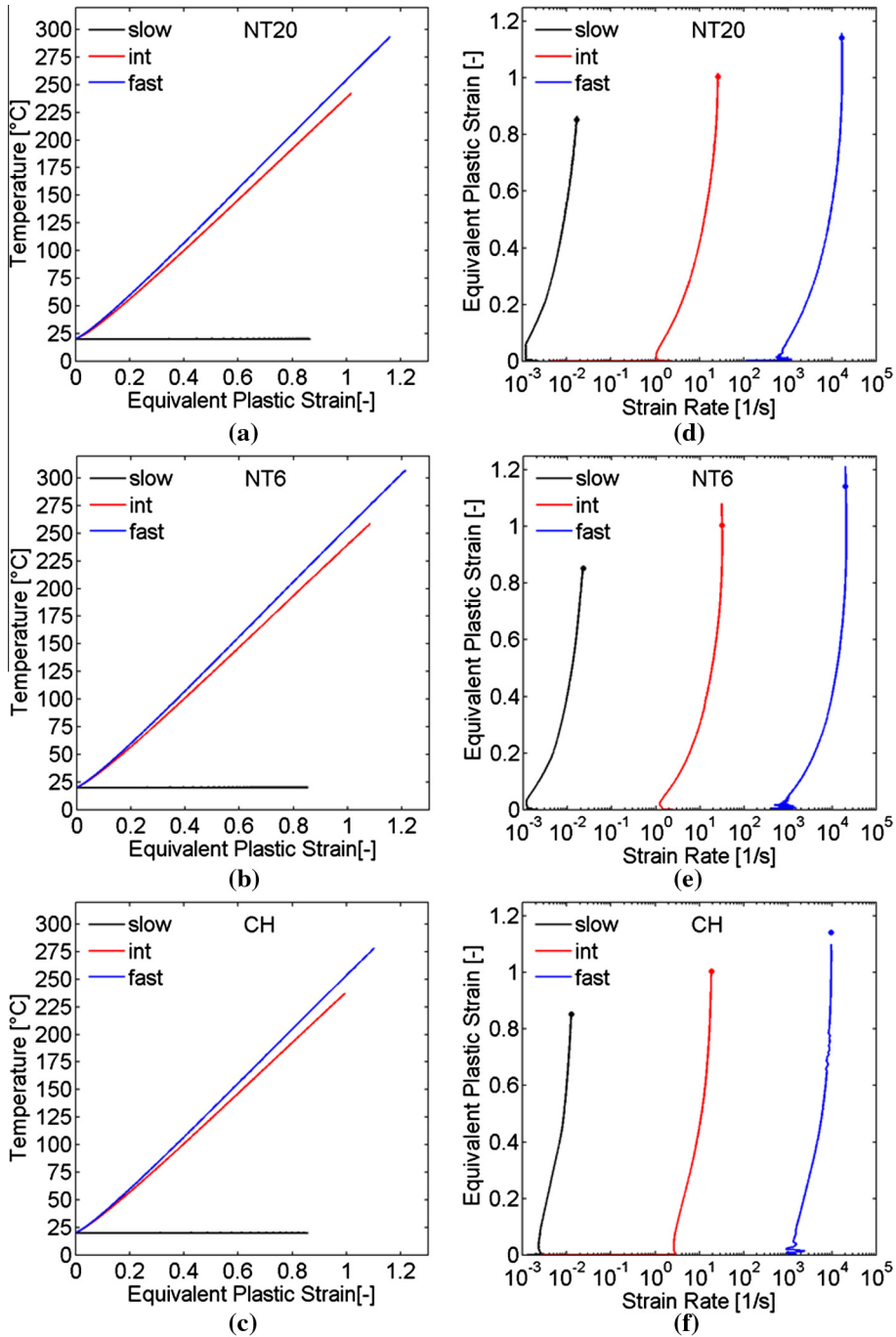


Fig. 13. Loading paths to fracture for NT20 (first line), NT6 (second line) and CH (third line) experiments on the DP590 steel. The left column figures show the predicted temperature as a function of the equivalent plastic strain, the right columns show the evolution of the equivalent plastic strain as a function of the strain rate. On each curve, the fracture strain predicted by the rate-modified Hosford–Coulomb model is indicated by a solid dot.

to confirm the particular parametric form of the function $\omega[\dot{\bar{\epsilon}}]$ given in Eq. (16). Our simulations for different forms revealed notable changes of the temperature at the specimen center for similar surface temperatures.

The longitudinal cross-sectional views of the specimens (right columns in Figs. 9–12) show the strain distribution throughout the necking phase. Up to the force maximum, plane stress conditions prevail and the strain distribution is approximately uniform along the thickness direction. Throughout necking, the deformation localizes within a small zone along the longitudinal axis and through-thickness gradients built up. The comparison of the cross-sectional views at the onset of fracture for different loading velocities clearly shows that the localization severity increases the higher the speed of loading.

The corresponding plots of the equivalent plastic strain versus the strain rate (right columns of Fig. 13) show that the strain rate increases by at least one order of magnitude after the onset of necking. For example, the pre-necking strain rate in the fast NT20 experiment is about $10^3/s$, while a strain rate of about $10^4/s$ is reached at the onset of fracture.

6. Rate-dependent Hosford–Coulomb fracture initiation model

6.1. Model formulation

An empirical extension of the Hosford–Coulomb fracture initiation model is proposed to account for the effect of strain rate on ductile fracture. The rate-independent Hosford–Coulomb model has been formulated assuming that ductile fracture is imminent with the onset of shear localization at the microscale (Mohr and Marcadet, submitted for publication). According to the Hosford–Coulomb model, fracture occurs at an equivalent plastic strain of $\bar{\epsilon}_f$ when the integral condition

$$\int_0^{\bar{\epsilon}_f} \frac{d\bar{\epsilon}_{pl}}{\bar{\epsilon}_f^{pr}[\eta, \theta]} = 1 \quad (22)$$

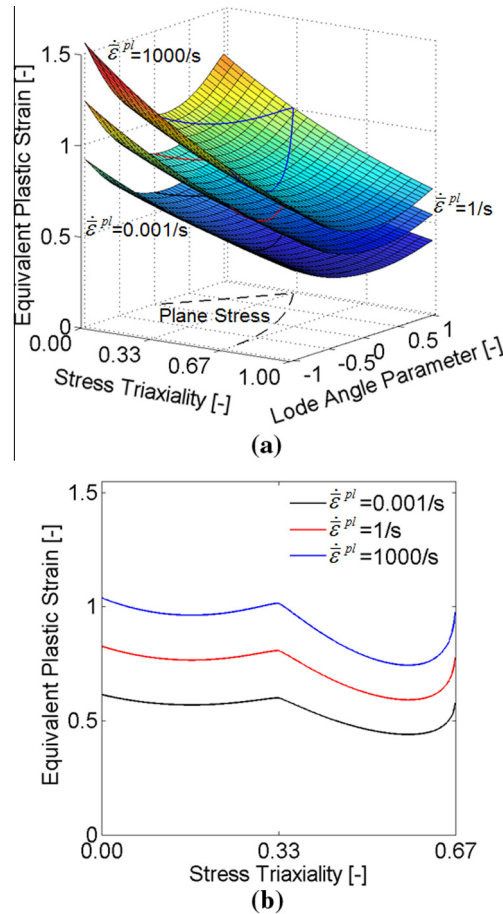


Fig. 14. (a) Illustration of the fracture surfaces (for proportional loading) for different strain rates in the space of stress triaxiality, Lode angle parameter and equivalent plastic strain. (b) Illustration of the corresponding fracture envelopes for plane stress conditions in the equivalent plastic strain versus stress triaxiality plane.

is fulfilled. In (22), the analytical expression for the strain to fracture for proportional loading is derived from the Hosford–Coulomb model in stress space,

$$\bar{\epsilon}_f^{pr}[\eta, \bar{\theta}] = b(1+c)^{\frac{1}{n}} \left(\left\{ \frac{1}{2} ((f_1 - f_2)^a + (f_2 - f_3)^a + (f_1 - f_3)^a) \right\}^{\frac{1}{a}} + c(2\eta + f_1 + f_3) \right)^{-\frac{1}{n}} \quad (23)$$

with the Lode angle parameter dependent trigonometric functions

$$f_1[\bar{\theta}] = \frac{2}{3} \cos \left[\frac{\pi}{6} (1 - \bar{\theta}) \right], \quad f_2[\bar{\theta}] = \frac{2}{3} \cos \left[\frac{\pi}{6} (3 + \bar{\theta}) \right] \quad \text{and} \quad f_3[\bar{\theta}] = -\frac{2}{3} \cos \left[\frac{\pi}{6} (1 + \bar{\theta}) \right].$$

The main model parameters are $\{a, b, c\}$. The Hosford exponent a controls the effect of the Lode angle parameter, while the friction coefficient c primarily controls the effect of the stress triaxiality on the strain to fracture. The model parameter b is a multiplier controlling the overall magnitude of the strain to fracture. It is defined such that it is equal to the strain to fracture for uniaxial tension (which is the same as that for equi-biaxial tension). The parameter n has only little effect on the calibrated model as a similar effect can also be achieved by adjusting the coefficients a and b (Eq. (23)). It originates from a power law type of isotropic hardening model. We chose $n = 0.1$ as it is a typical hardening exponent for metals. However, it is reemphasized that a different value could also be chosen without changing the reported results.

In loose analogy with the Johnson and Cook (1985) fracture model, the effect of strain rate is incorporated into the Hosford–Coulomb model through the coefficient b ,

$$b = \begin{cases} b_0 & \text{for } \dot{\epsilon}_p < \dot{\epsilon}_0 \\ b_0(1 + \gamma \ln \frac{\dot{\epsilon}_p}{\dot{\epsilon}_0}) & \text{for } \dot{\epsilon}_p \geq \dot{\epsilon}_0 \end{cases} \quad (24)$$

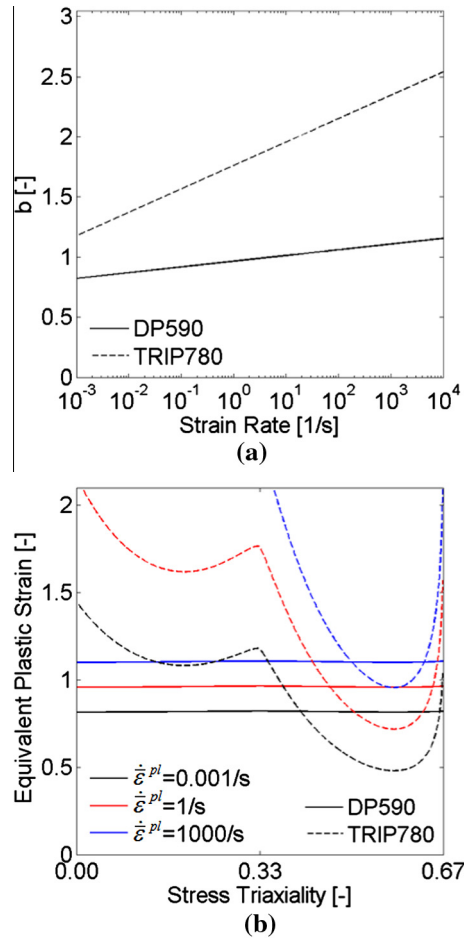


Fig. 15. (a) Parameter b of the rate-modified Hosford–Coulomb model as a function of the strain rate as calibrated for the DP590 and TRIP780 steels. (b) Plot of the calibrated fracture envelopes for different strain rates.

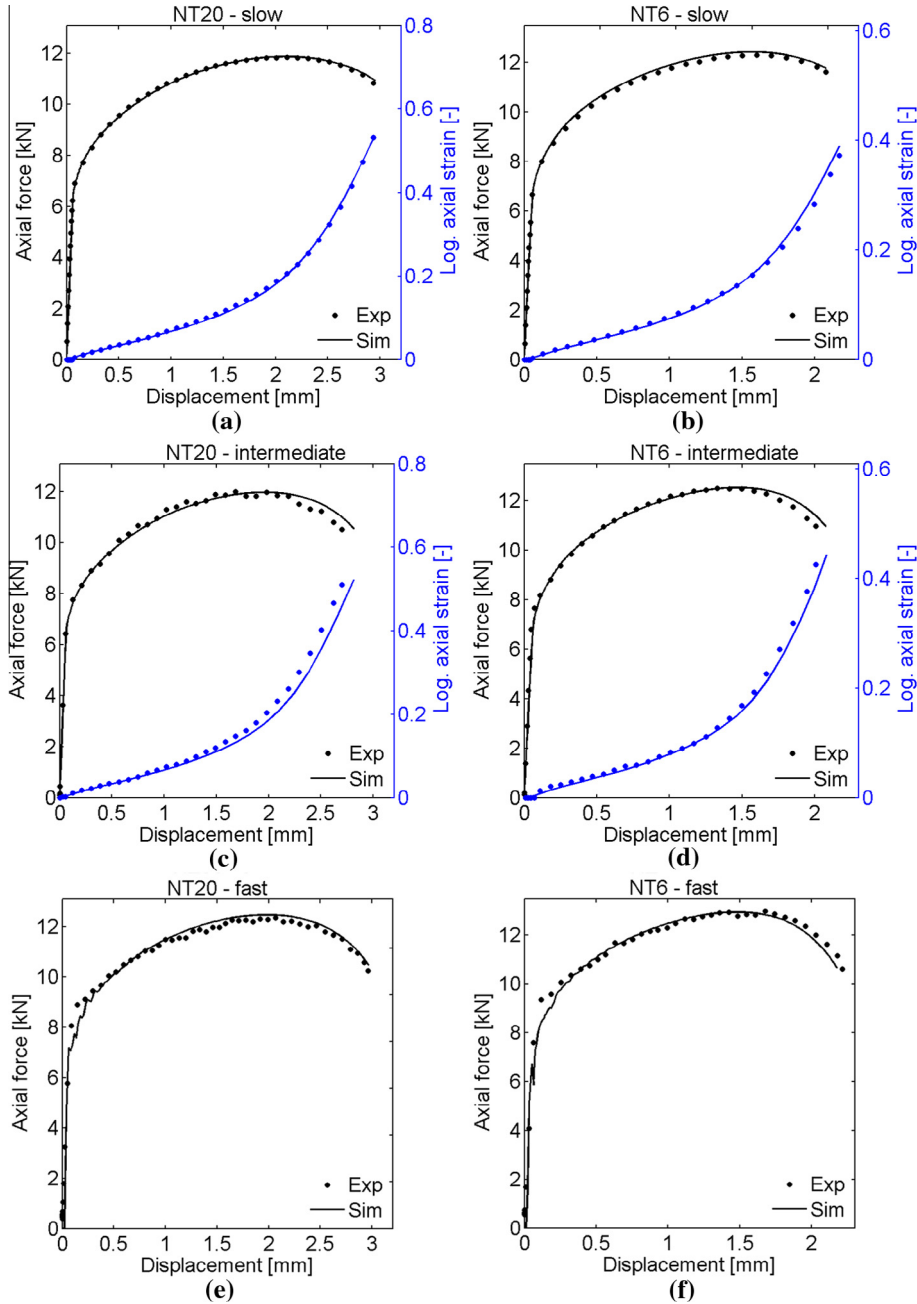


Fig. 16. Force–displacement and surface strain versus displacement curves from the NT20 (left column) and NT6 (right column) experiments and simulations for the TRIP780 steel. No local strain measurements were available for fast loading (Dunand, 2013).

$b_0 > 0$ is the strain to fracture for uniaxial tension at low strain rates, while the model parameter $\gamma \geq 0$ controls the strain rate sensitivity of the fracture initiation model. Fig. 14a illustrates the effect of the strain rate on the resulting “fracture surface” in the space of equivalent plastic strain, stress triaxiality and Lode angle parameter. The corresponding fracture envelopes for plane stress loading are shown in Fig. 14b.

6.2. Model validation for DP590 steel

The calibration of the fracture initiation model requires a characterization of the effects of stress state (parameters a and c) and strain rate (parameters b_0 and γ). Based on the results of Mohr and Marcadet (submitted for publication) on the stress state sensitivity of the present DP590 steel under static loading, the parameters $a = 1.97$ and $c = 0$ are chosen to account for

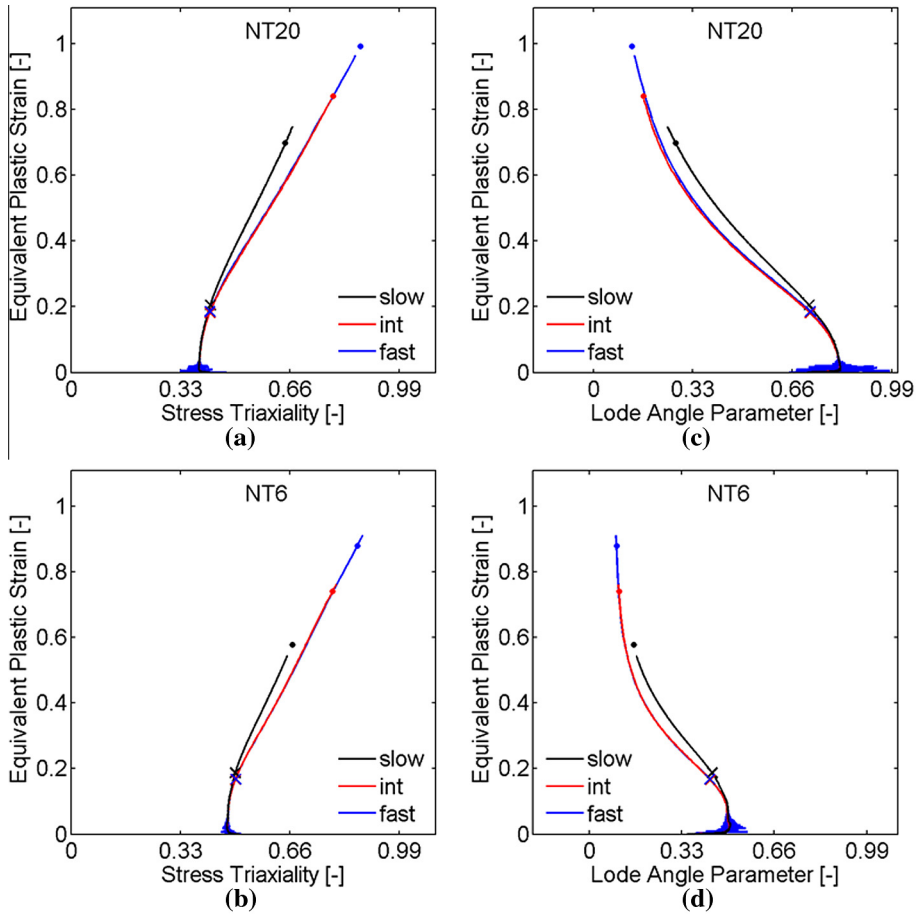


Fig. 17. Loading paths to fracture for NT20 (first line) and NT6 (second line) experiments on the TRIP780 steel. The left column figures show the equivalent plastic strain as a function of the stress triaxiality, the right columns show the evolution of the equivalent plastic strain as a function of the Lode angle. On each curve, the fracture strain predicted by the rate-modified Hosford–Coulomb model is indicated by a solid dot; the instant of maximum force is highlighted by a cross symbol.

the mild influence of the Lode angle on the fracture strain. The parameters controlling the rate dependency are then calibrated based on the loading paths to fracture for the slow, intermediate and fast NT20 experiments. The solid dots in Figs. 12 and 13 highlight the instants of onset of fracture predicted by the above model using the parameters $b_0 = 0.82$ and $\gamma = 0.025$. Good agreement of the predicted instants of fracture (solid dots) and those identified experimentally (respective end points of the solid loading paths) is not only observed for all NT20 experiments (used for calibration), but also for all NT6 and CH experiments, with a maximum relative error in the predicted strain to fracture of 6.5%. Fig. 15a shows the evolution of b versus $\dot{\epsilon}_p$, while Fig. 15b shows the fracture envelope for plane stress conditions.

6.3. Model validation for TRIP780 steel

The proposed plasticity and fracture models are also applied to a 1.4 mm thick TRIP780 steel. Dunand (2013) performed NT6 and NT20 experiments for slow, intermediate and fast loading conditions on this material using the same experimental procedures as those outlined above. The stress state sensitivity parameters $a = 1.22$ and $c = 0.11$ had been identified for this type of material by Mohr and Marcadet (submitted for publication) based on the experimental results from butterfly, CH, NT and punch specimens.² Following the same procedure as for the DP590 material, the parameters of the above rate-dependent plasticity model have been identified (Table 4). The corresponding force–displacement curves are shown in Fig. 16. Note the good agreement of the numerical simulations with the experiments in the post-necking range.

The extracted loading paths to fracture for the NT6 and NT20 experiments are shown in Figs. 17 and 18. The strain to fracture increases from 0.58 to 0.93 when increasing the speed of loading from $7.6 \cdot 10^{-6}$ m/s to 4.2 m/s in a NT6 experiment.

² The results for the TRIP780 steel reported in earlier publications were from a different batch of sheets which is the main reason for slight differences in the reported material properties for static loading.

As for the DP590 steel, the force maximum is already reached at an equivalent plastic strain of about 0.2, i.e. a substantial portion of the total deformation is accumulated in the post-necking range. The b versus $\dot{\epsilon}_p$ curve of the rate-dependent fracture model, as calibrated based on the NT20 and punch results, is shown as a dashed curve in Fig. 15a ($b_0 = 1.27$ and $\gamma = 0.066$). The corresponding model predictions of the instant of onset of fracture (solid dots in Fig. 17) agree well with the experiments (end of the loading paths). Good agreement is also observed for NT6, with a maximum error of 6.4% in the strain to fracture for the slow experiment. The complete fracture envelope for the TRIP780 is depicted with a dashed line in Fig. 15b.

6.4. Discussion

The proposed rate-dependent Hosford–Coulomb model integrates the effect of the strain rate on an empirical basis. This approach shows reasonable results for the two advanced high strength steels considered here. However, it does not differentiate between the effects of temperature and strain rate. It appears to be impossible to perform high strain rate experiments under isothermal conditions (except for multiple loadings, e.g. Nemat-Nasser and Isaacs, 1997). Static experiments at different temperatures would be an option, but other phenomena such as dynamic strain aging are expected to come into play (e.g. Queiroz et al., 2012; Chaboche et al., 2013). It is therefore expected that shear localization analysis of unit cell models (see Barsoum and Faleskog (2007) and Dunand and Mohr (2014)) with a temperature- and rate-dependent matrix material will provide an answer to the fundamental question of the respective effects of strain rate and temperature on the onset of ductile fracture.

In engineering practice, the present model is probably sufficient since the temperature fields are seldom calculated in a rigorous manner. Moreover, it is worth noting that the simulation results are mesh-size dependent. Dunand and Mohr (2010) showed an example of numerical simulations with a rate- and temperature independent plasticity model that do converge with respect to the element size even after the onset of necking precedes fracture. In other words, the stress-state dependent

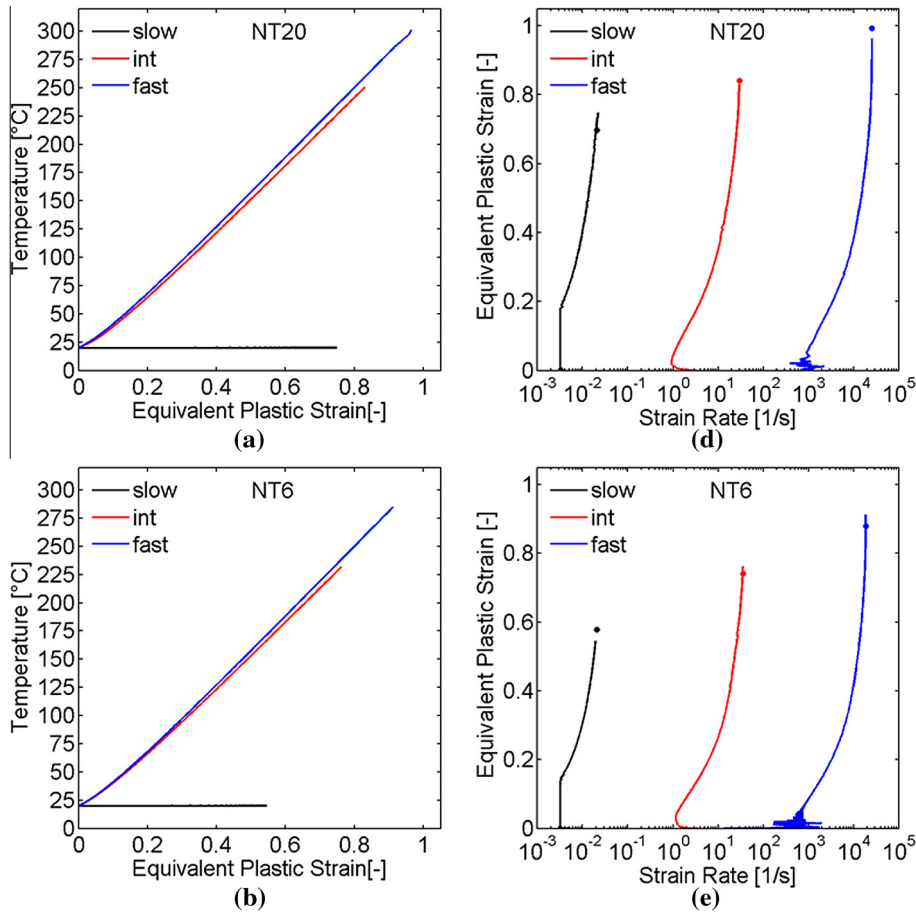


Fig. 18. Loading paths to fracture for NT20 (first line) and NT6 (second line) experiments on the TRIP780 steel. The left column figures show the predicted temperature as a function of the equivalent plastic strain, the right columns show the evolution of the equivalent plastic strain as a function of the strain rate. On each curve, the fracture strain predicted by the rate-modified Hosford–Coulomb model is indicated by a solid dot.

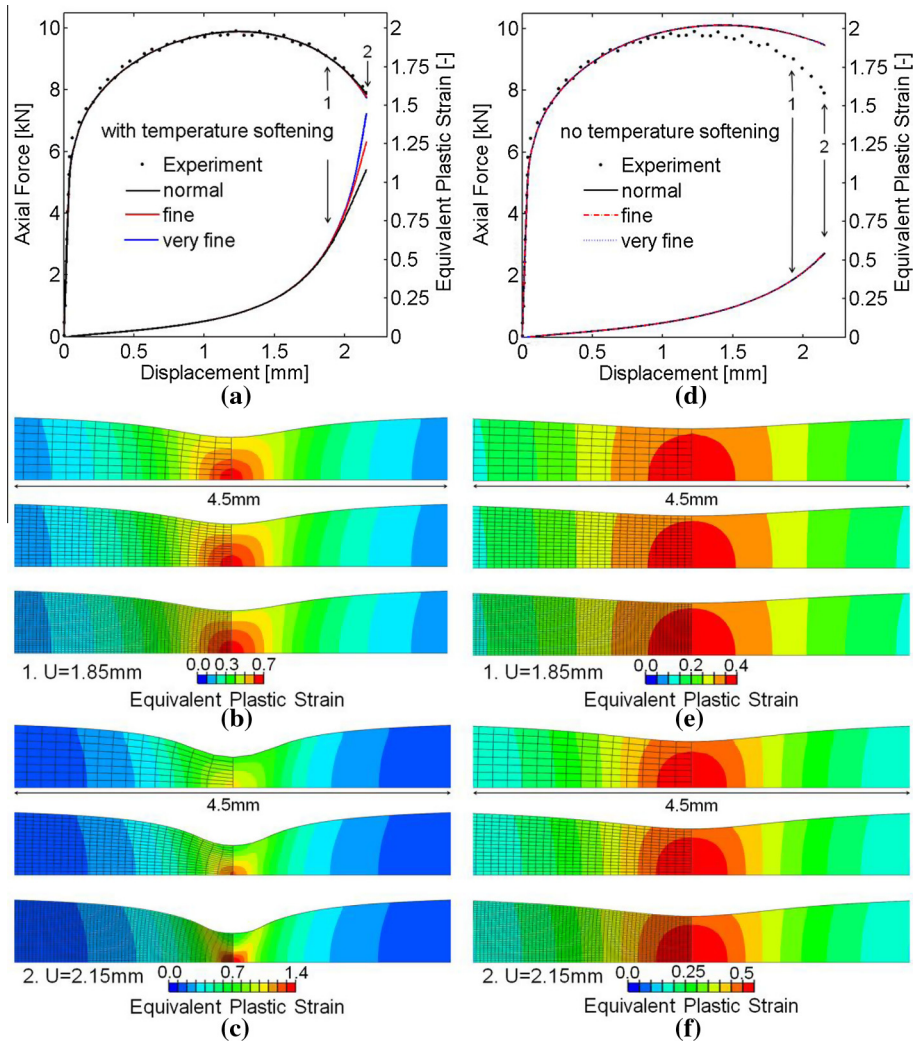


Fig. 19. Illustration of the mesh size dependency of the simulation results (intermediate speed NT6 simulations for DP590). The left column shows the results for active thermal softening (i.e. the full rate- and temperature dependent constitutive model). The right column shows the results from the same simulations with inactive thermal softening. The “normal” mesh features 8 elements through half the sheet thickness, the “fine” mesh has 16, and the “very fine” mesh has 32.

parameters of the fracture initiation model could be determined in a rigorous manner. However, this convergence appears to break down when thermal softening comes into play. This is briefly illustrated in Fig. 19 where we present the results from numerical simulations of the intermediate velocity NT6 experiments with active thermal softening (left column) and inactive thermal softening (right column). Observe that the simulation results converge with respect to the mesh size when thermal softening is inactive. The reported fracture model parameters controlling the rate dependency are therefore expected to be mesh size dependent. Different techniques are available to address this issue from a theoretical and computational point of view (e.g. see review by Bazant and Jirásek (2002)), but further research is needed in experimental mechanics to measure the corresponding fields (temperature, strain and strain rate) at the location of onset of fracture.

7. Conclusions

Static and dynamic experiments are performed on specimens extracted from advanced high strength steel sheets to investigate the effect of loading speed on fracture initiation. The experimental results for DP590 and TRIP780 steel for notched tension and tension with a central hole indicate that the ductility increases by as much as 60% when increasing the loading speed from $7.6 \cdot 10^{-6}$ m/s to 4.2 m/s.

A plasticity model is proposed which couples a Johnson–Cook type of rate- and temperature dependency with a combined Swift–Voce strain hardening law. The model treats the temperature as an internal variable and can therefore account for the

effect of thermal softening in an approximate manner without solving the thermal field equations. Reasonable agreement of the simulation and experimental results is observed for different specimen geometries, different loading velocities and different materials. In particular, the model is able to provide an accurate description of the post-necking phase, where most of the deformation accumulates. The results from the numerical simulations are subsequently used to extract the histories of stress triaxiality, Lode angle, equivalent plastic strain, strain rate and temperature at the location of fracture initiation.

Mohr and Marcadet (submitted for publication) had proposed a Hosford–Coulomb model to describe the stress state sensitivity of fracture initiation of the present materials at low strain rates. Here, an empirical extension of the Hosford–Coulomb model is proposed to account for the effect of strain rate on fracture initiation. The two model parameters are calibrated for notched tension (NT20), before validating the model's predictive capabilities for all remaining static and dynamic experiments.

Conflict of interest

No conflict of interest.

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